

VIBROACOUSTICS - ASSIGNMENT 3

Sound Radiation from plate-like structures

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Introduction

This third part focuses on the analytical evaluation of the acoustic field radiated by a vibrating plate. The plate, is assumed to be mounted in an infinite rigid baffle and excited by a force transmitted through a string. The purpose of this analysis is to estimate the levels of sound pressure perceived by virtual receivers located in a hemispherical domain, based on the vibration velocity of the plate in the frequency range [1–2000 Hz]. The results include the estimation of the complex acoustic pressure at arbitrary locations, the computation of the Sound Pressure Level (SPL) in one-third octave bands for ten microphone positions specified by the EN 60704-1:2021 standard, and the graphical visualization of the SPL distribution at key frequencies.

1 Single-point sound radiation prediction

In this section, we analytically estimate the sound pressure radiated by the vibrating plate (previously characterized in Lab02) toward a generic observation point in space. Assuming that the plate is flush-mounted in an infinite rigid baffle, the radiated acoustic field can be described using a simplified form of the Kirchhoff–Helmholtz integral equation, valid in the frequency domain:

$$p(\vec{r}) = j\omega\rho_0 \iint_S v(x_0, y_0) G(\vec{r} | x_0, y_0, 0) dx_0 dy_0 \quad (1)$$

In free-field conditions, the Green's function G for a monopole source located on a rigid baffle is:

$$G(\vec{r} | x_0, y_0, 0) = \frac{e^{-jkr}}{2\pi r} \quad (2)$$

where:

- $\omega = 2\pi f$ is the angular frequency (in this case, $f = 440$ Hz);
- $k = \omega/c$ is the acoustic wavenumber;
- $r = \sqrt{(x - x_0)^2 + (y - y_0)^2 + z^2}$ is the distance from the plate element (x_0, y_0) to the receiver;
- $\rho_0 = 1.2$ kg/m³ is the air density;
- $c = 343$ m/s is the speed of sound in air;
- $v(x_0, y_0)$ is the local normal velocity of the plate, assumed known from modal superposition.

The pressure $p(\vec{r})$ represents the cumulative acoustic effect of all infinitesimal sources distributed across the plate surface, each radiating with local velocity $v(x_0, y_0)$.

For the numerical computation, the plate surface is discretized over a rectangular mesh using 200 points along its length a and 100 points along its width b .

As an example, for a receiver located at coordinates $(x, y, z) = (1.5, 1.2, 3)$, the resulting complex sound pressure computed from the velocity distribution is approximately:

$$p = 0.138 + 0.227j \quad (\text{Pa})$$

This method is derived under the far-field approximation, where the receiver is located at a distance much greater than the dimensions of the vibrating plate. In this configuration, the pressure field behaves as a spherical wave and decays proportionally to $1/r$, as expected from the free-field Green's function for a distributed source. Although the method can still provide reasonable estimates in the near field, its accuracy improves with distance, especially when the variation of r across the plate surface becomes negligible (Rayleigh condition).

2 Microphone array and SPL estimation

In accordance with the EN 60704-1:2021 standard for acoustic measurements, ten microphone positions are defined on the surface of a virtual hemisphere surrounding the sound source. The radius of the hemisphere is arbitrary; for this study, a value of $R = 4$ m is selected, placing the center of the hemisphere at the geometrical center of the vibrating plate. This configuration ensures uniform spatial sampling of the radiated field and allows for robust estimation of spatial variations in sound pressure.

To evaluate the perceived sound intensity, we compute the Sound Pressure Level (SPL) in one-third octave bands for each microphone. The SPL for band i is calculated by integrating the squared pressure values $|p_j|^2$ over the frequency range $[f_{i,\min}, f_{i,\max}]$ corresponding to that band:

$$L_{p,i} = 10 \log_{10} \left(\frac{\sum_{j=f_{i,\min}}^{f_{i,\max}} |p_j|^2}{p_{\text{ref}}^2} \right) \quad \text{with } p_{\text{ref}} = 2 \times 10^{-5} \text{ Pa} \quad (3)$$

This expression accumulates the energy of all spectral components falling within the third-octave band limits and normalizes it with respect to the reference pressure of human hearing threshold. The logarithmic scaling expresses the result in decibels (dB SPL). The result provides a frequency-averaged measure of the sound energy perceived by each microphone in the selected band.

No.	x/R	y/R	z/R
1	-0,99	0	0,15
2	0,50	-0,86	0,15
3	0,50	0,86	0,15
4	-0,45	0,77	0,45
5	-0,45	-0,77	0,45
6	0,89	0	0,45
7	0,33	0,57	0,75
8	-0,66	0	0,75
9	0,33	-0,57	0,75
10	0	0	1,0

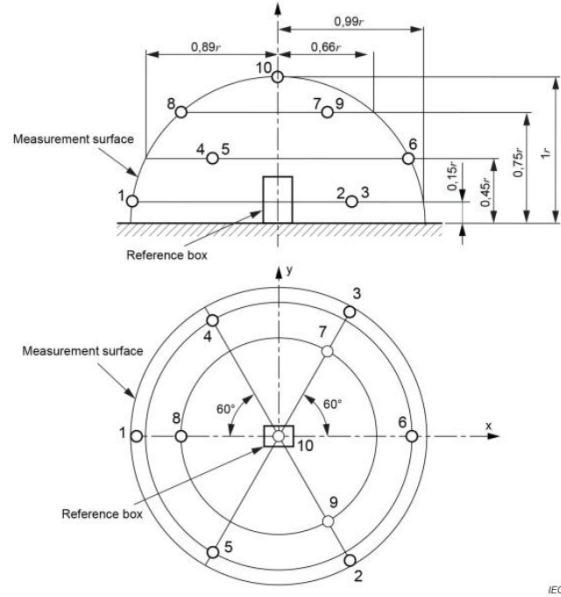


Figure 1: Microphones positions on the hemispherical array.

Figure 2: Top view of measurement positions.

3 SPL 3D Representations

To evaluate the directional characteristics of the acoustic radiation, we compute and visualize the Sound Pressure Level (SPL) in one-third octave bands centered at 250 Hz and 500 Hz. These bands are defined respectively by the frequency intervals [225–283] Hz and [440–565] Hz.

The SPL values at the ten virtual microphones are interpolated over a hemispherical surface of radius 4 m using the function `sphere_geom_VAV.m`, which provides the geometry required to generate the 3D surface plots. The interpolation is based on the spatial distribution of the measured SPL values, and allows for a smooth graphical representation of the acoustic field.

Theoretical analysis based on the physical properties of the plate allows us to compute its critical frequency, above which efficient radiation of bending waves becomes possible:

$$B = \frac{Es^3}{12(1-\nu^2)} = 519.52 \text{ Pa} \quad , \quad f_c = \frac{1}{2\pi} \sqrt{\frac{B}{\rho_0 s}} = 1486 \text{ Hz}$$

Since both 250 Hz and 500 Hz are well below the critical frequency, the plate operates in a radiation regime dominated by edge and corner modes. This is due to the fact that, at these frequencies, the acoustic wavelength exceeds at least one or both dimensions of the plate.

At 250 Hz, the acoustic wavelength is:

$$\lambda_{250} = \frac{c}{f} = \frac{343}{250} \approx 1.37 \text{ m},$$

which is significantly larger than both the plate length ($\lambda_x = 0.8 \text{ m}$) and width ($\lambda_y = 0.5 \text{ m}$). Under this condition, the entire plate tends to vibrate in phase, resulting in omnidirectional radiation with limited spatial variability. This behavior is consistent with a monopole-like source and leads to a relatively uniform SPL distribution with variations around 2–3 dB across the hemisphere.

At 500 Hz, the acoustic wavelength decreases to:

$$\lambda_{500} = \frac{343}{500} \approx 0.686 \text{ m},$$

which remains larger than the plate width but becomes smaller than its length. In this regime, modal cancellation effects become more significant along the longer axis, and radiation becomes more directional. The SPL field exhibits stronger spatial variations (up to 8–10 dB), with localized lobes and nulls corresponding to constructive and destructive interference patterns.

As the frequency increases, these effects intensify: radiation efficiency improves, spatial directivity increases, and the overall SPL rises. The figures below confirm these theoretical predictions, showing a more uniform and diffuse field at 250 Hz and a sharper, more complex pattern at 500 Hz.

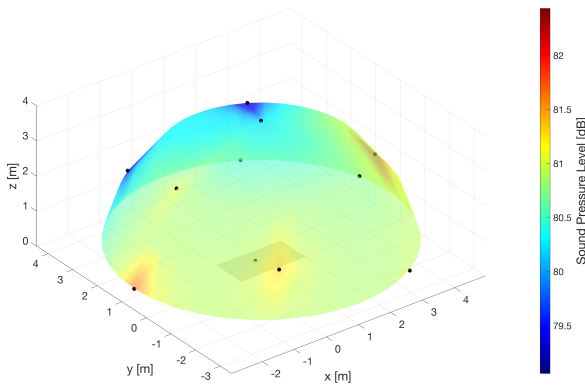


Figure 3: SPL distribution at 250 Hz (One-Third Octave Band)

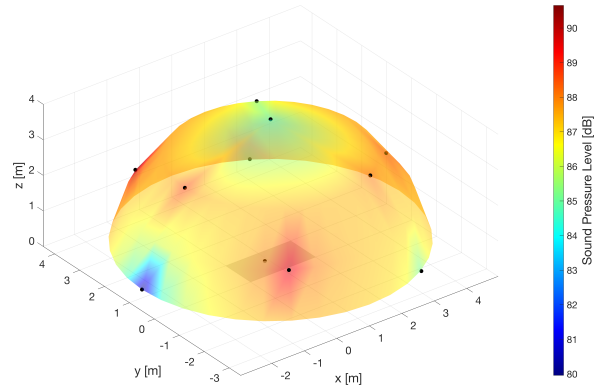


Figure 4: SPL distribution at 500 Hz (One-Third Octave Band)

Finally, the SPL values used for validation are reported in Table 1, showing consistent trends with theoretical expectations and confirming the accuracy of the implemented model.

Table 1: SPL values at 250 Hz and 500 Hz measured by 10 virtual microphones.

Microphone	250 Hz [dB]	500 Hz [dB]
Micro 1	81.73	80.02
Micro 2	80.94	85.01
Micro 3	81.06	90.58
Micro 4	79.10	90.39
Micro 5	81.19	89.46
Micro 6	82.43	89.68
Micro 7	78.98	85.51
Micro 8	81.07	88.53
Micro 9	80.67	88.00
Micro 10	80.23	84.37